AURA 2026 Data Quality & Exploratory Analysis Report

Dataset description: 286,696 valid observations were parsed from AY_Project_Data_1786812502784.xlsx. The workbook contains split date/time columns, repeated metadata/header rows, and missing sentinel values.
Cleaning methodology: 3,121 non-observation rows were excluded from the analysis frame. Exact duplicate rows: 0. Duplicate timestamps: 0; conflicting timestamp groups: 0. The -9999 sentinel was preserved as missing, not converted to zero.
The observation dates were calendar-shifted from 2022 to 2026 for the project’s forecasting timeline while preserving the original month, day and time structure.
Chronological split: 202,965 records from February-August 2026 were eligible for training; 74,021 records from September 2026 onward were held out. No holdout rows were used for model fitting, feature fitting, or validation.
Model: Chronological multivariate linear regression targeting refractivity. Validation used the final training month as a chronological validation period. Metrics: MAE=26.136164, RMSE=31.693458, R2=-4.695695, bias=-23.726993.
Limitations: future environmental predictors are represented by training-period hour-of-day climatology because the project prohibits external data. The forecast is therefore an explainable baseline for demonstration and should not be described as a full operational weather forecast.
April vs July means: {"April 2026": {"temperature": 25.1325, "relativeHumidity": 52.0312, "pressure": 968.9413, "heatIndex": 24.3313, "refractivity": 321.8232}, "July 2026": {"temperature": 26.6256, "relativeHumidity": 77.897, "pressure": 970.8459, "heatIndex": 24.668, "refractivity": 363.7233}}. Correlations are descriptive associations only and do not prove causation.